Huan Phan-Van

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EDUCATION:

Mechanical Engineering, Ph.D. — *Fall 2020 – Fall 2025*
Southern Illinois University Carbondale (SIUC), in Robotics — **Edwardsville, IL**

Mechanical Engineering, M.S. — *Fall 2018 - Summer 2020*
Southern Illinois University Edwardsville (SIUE), in Robotics — **Edwardsville, IL**

Mechanical Engineering, B.E. — *Sept. 2009 - May 2013*
HCMC University of Technology and Education (HCMC UTE) — **Ho Chi Minh, Vietnam**

INDUSTRY EXPERIENCE:

TuSimple – Autonomous Trucking — **Tucson, AZ**
Sensor Hardware Engineer Intern — *May 2022 – Aug. 2022*

- Evaluated **LiDAR, Camera,** and **IMU** performance for level 4 Autonomous Truck application.
- **LiDAR:** evaluated the **near-range LiDAR** Cepton Nova for FOV-Resolution, Detection-Range, Range Accuracy, Small Objects Detection, Mirror Noise, Blooming Noise, and Scanning Pattern.
- **IMU:** tested the performance of an IMU by using **Allan Variance**.
- **Camera**: used **NEDT** to test the thermal sensitivity of the camera, used **Imatest** to evaluate Camera ISO Sensitivity and Exposure Index.
- Used **MATLAB**, **Python**, and **CloudCompare** to do point cloud analysis, able to analyze Sunny days and Rainy days for the **mid-range LiDAR** M1 performance.
- Used **ROS, CAN**, and **Serial** to collect, visualize and analyze sensor data performance evaluation.

Kawahara Co., Ltd. — **Nagoya, Japan**
Mechanical Engineer — *Oct 2016–Jun 2017*

- Led quality control for precision automotive components; maintained CNC machining systems.

Sakae Seiko Co., Ltd. — **Obu, Japan**
Mechanical Engineer — *June 2015–Sep 2016*

- Designed and programmed machining paths using SolidWorks and Mastercam.

RESEARCH & PROJECT EXPERIENCE:

Time-Delay Systems and Feedback Control — SIUE (2020–Present)

- Developed structured invariant-subspace methods for stability and control of time-delay systems.
- Designed Lyapunov–Krasovskii-based controllers validated in MATLAB/Simulink.
- Built six omnidirectional and two agricultural robots integrating LiDAR, IMU, and camera sensors.

Autonomous Robotics Development

- Implemented trajectory tracking and backstepping control under uncertainty.
- Developed sensor-fusion algorithms and experimental validation for mobile-robot platforms.

PUBLICATIONS:

- H. Phan-Van and K. Gu (2024), "Structured Invariant Subspace and Decomposition of Systems with Time Delays and Uncertainties," *International Journal of Robust and Nonlinear Control.*
- K. Gu and H. Phan-Van (2023), "External Direct Sum Invariant Subspace and Decomposition of Coupled Differential–Difference Equations," *IEEE Transactions on Automatic Control.*
- H. Phan-Van, H. Ferdowsi, and N. Lotfi (2020), "Trajectory Control of Omnidirectional Mobile Robots Considering Low-Level Actuator Dynamics," *28th Mediterranean Conference on Control and Automation (MED).*
- N. Lotfi, J. A. Novosad, and H. Phan-Van (2019), "A Multidisciplinary Course and Laboratory Platform for Teaching the Fundamentals of Advanced Autonomous Vehicles," *ASEE Annual Conference & Exposition.*

TEACHING EXPERIENCE:

Southern Illinois University Edwardsville (SIUE) — **Edwardsville, IL**
Instructor — *Aug. 2020 - Present*

- Courses: (ME-354) Numerical Simulation (MATLAB), (MRE-358) Introduction to Mechatronics (LabVIEW), (ME-320) **Sensors and Actuators, (**ME-563) **Optimal Control**
- Applied MATLAB, LabVIEW, and ROS to teaching materials.

HONORS & AWARDS:

- **The university's 2020 Outstanding Teaching Assistant Award** — **SIUE, Edwardsville, IL**
- Research Grants for Graduate Students (RGGS)